RFID-Integrated

Child Safety System

Where Innovation Meets Child Protection

Team Details

Ideators involved:

Pratik Tiwari

Vidhi Umap

Vedant Sikariya

Institution:

Ramdeobaba University, Nagpur

Superviser:

Dr. R. R. Khandelwal

Problem Statement

Why is this project needed?

- Ensuring child safety inside school premises.
- Preventing unauthorized movement outside safe zones.
- Providing real-time tracking for parents and school authorities.

Project Objective

- Track and monitor students in real-time using RFID & GPS.
- Record attendance automatically via RFID scanning.
- Alert authorities and parents if a child moves outside a designated area.

System Overview

Main Components and Flow:

- RFID Attendance System (Arduino + RFID Reader)
- Live GPS Tracking System (ESP32 + GPS + GSM)
- Firebase Cloud Storage (to store location data)
- Web Dashboard/App (to view child’s location)

System Overview (Flow Diagram)

Scanning Bench

Arduino UNO

RC522

ESP32

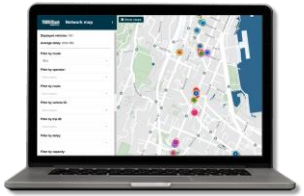
Tracking Device

NEO-6M GPS

SIM800L GSM

Display

Firebase



Components Used

1st Microcontroller (Arduino) - RFID Attendance System

- Arduino Uno – Microcontroller for RFID processing
- RC522 RFID Module – Scans student ID card
- LED & Buzzer – Indicates successful scanning
- Power Source – USB power supply

2nd Microcontroller (ESP32) - GPS Tracker

- ESP32 – Handles GPS tracking & communication
- NEO-6M GPS Module – Fetches real-time location
- SIM800L GSM Module – Sends location data to the cloud
- Li-Po Battery – Powers the GPS tracker

Working Process

- **Step 1:** Attendance System (RFID Scanning Bench)Student scans their RFID-enabled school ID card. Arduino logs attendance and triggers a buzzer & LED confirmation.
- **Step 2:** Real-Time GPS Tracking (ESP32 + GPS + GSM)GPS module tracks child's location continuously. ESP32 processes the data & sends it to Firebase via GSM.
- **Step 3:** Web/App-Based Monitoring Parents & school authorities view live location via a web dashboard or mobile app. If a child moves outside the safe zone, an alert is sent.

Key Features

- RFID-based Attendance System
- Live GPS Tracking via GSM
- Safe-Zone Monitoring & Alerts
- Web Dashboard for Monitoring
- Low-Power, Portable System for Easy Use

Expected Challenges & Solutions

Challenges	Solutions
Powering GPS tracker	Using Li-Po battery for portability
GSM network issues	Choosing strong network SIM (Jio, Airtel, Vi, BSNL, etc.)
Accuracy of GPS	Using high-quality GPS module (NEO-6M)

Cost Analysis

Components	Quantity	Price
ESP32	1	₹600
Arduino UNO	1	₹500
RFID Reader (RC522)	1	₹250
RFID Cards	5	₹200
GPS Module (Neo-6M)	1	₹700
GSM Module (SIM800L)	1	₹600
LEDs & Buzzer	5mm LED & 1 Buzzer	₹100
Jumping Wires & Breadbord	1 set	₹200
Li-Po Battery	1	₹300
		Total = ₹3500

Future Enhancements

- 🚀 Improving GPS Accuracy with better sensors.
- 📡 Implementing LoRa/Wi-Fi for Wireless Communication.
- 📱 Developing a Mobile App for Parents.
- 🔋 Exploring Solar-Powered GPS Trackers

Conclusion

- Successfully integrates RFID, GPS & GSM for child safety.
- Provides real-time tracking to parents & school authorities.
- Ensures children stay within a safe zone inside school premises.

THANKS